

TONY BAKER

Distinguished Engineer | AI & Agentic Development | Software Architect | Senior Individual Contributor

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PROFILE

Distinguished Engineer and software architect with 26+ years of experience building commercial software products, modernising legacy platforms and turning complex business problems into reliable software systems.

Seeking senior individual contributor roles that stay close to the technology: designing systems, writing code, building prototypes, integrating AI into products and helping engineering teams adopt practical AI workflows.

Known for combining deep technical judgement with commercial awareness, strong interpersonal skills and the ability to work effectively across global teams and time zones.

TARGET ROLES

Distinguished Engineer | AI Engineering Lead | AI Architect | AI Product Engineer

CORE SKILLS

AI & agents: AI-assisted development | agentic workflows | RAG systems | vector databases | custom agents | MCP | Microsoft Foundry / Azure AI Foundry | OpenAI APIs | Azure OpenAI | Claude Code | Cursor

Architecture: software architecture | distributed systems | microservices | event-driven architecture | forecasting platforms | SaaS | critical infrastructure

Engineering: C# | .NET | TypeScript | React | SQL Server | Oracle | RabbitMQ | Azure | AWS | Kubernetes | Azure DevOps | CI/CD

Enablement: engineering productivity | technical strategy | mentoring | developer enablement | cross-functional collaboration | global teams

SELECTED IMPACT

- Built APAC operations from scratch into a high-performing 10-person multidisciplinary team spanning international business development, sales, support, engineering, forecasting analysis and operations.
- Built and evolved TESLAWeb2 as primary developer from 2010 to 2023; a flagship forecasting platform used by TESLA Forecasting customers globally and still in customer use during transition to the next platform generation.
- Led AI adoption initiatives resulting in 100% AI-assisted development adoption across Forecasting engineering teams in APAC and the United States, supported by workshops, mentoring and practical implementation guidance.
- Architected and delivered a suite of forecasting microservices that automated model setup, scheduling, aggregation, ensemble processing and legacy conversion, reducing complex multi-team configuration work from several hours to approximately 10 minutes via analyst self-service.
- Reduced ensemble forecasting solve time from approximately 25 minutes to under 1 minute and replaced thousands of lines of configuration with a small automatically generated configuration file.
- Built a proof-of-concept RAG chatbot for RTO Insider by converting approximately 20,000 energy news articles into vector-searchable content and grounding an agent chatbot in that knowledge base with query rewriting for quality and safety.
- Delivered measurable performance improvements, including reducing a slow API path from 2m20s to 25s, reducing selected workflow processing times by 80-99%, and lowering database CPU demand by approximately 10-15% on a large production system.
- Built a utility to generate and migrate historical forecast data into the Yes Energy platform, including a functional run producing 38,800 files for MISO Total.

PROFESSIONAL EXPERIENCE

Distinguished Engineer | Yes Energy

May 2024 - Aug 2024 | Auckland, New Zealand / Remote

- Operate as a senior hands-on individual contributor focused on AI adoption, forecasting platform modernisation, microservice architecture, technical strategy and cross-team engineering enablement.
- Led AI adoption across Forecasting engineering teams in APAC and the United States, achieving 100% AI-assisted development adoption through hands-on coaching, workshops, practical examples and guardrails.
- Designed and built a production forecasting automation platform that consolidated model execution, scheduling, aggregation and ensemble processing, reducing manual handoffs, configuration risk and dependency on reliability/SysOps teams.
- Reduced complex forecasting setup from hours of analyst, engineering and operations coordination to approximately 10 minutes of analyst configuration via one web page.
- Created Azure DevOps CI/CD pipelines and maintainable deployment patterns for forecasting services and tools.
- Built a tool to support migration of historical forecasts into the Yes Energy platform, addressing a known sales and customer value gap caused by limited historical forecast availability.
- Built AI proof-of-concepts including RAG systems, custom agents, vector database integrations and AI chat interfaces using OpenAI, Azure OpenAI, Claude/Cursor and Microsoft Foundry / Azure AI Foundry learning projects.
- Converted approximately 20,000 RTO Insider energy news articles into vector-searchable data and built a grounded chatbot with query rewriting and safety/quality controls.
- Optimised core services and shared library code, including reducing processing times by 80-99% in selected workflows and lowering database CPU demand on production systems.
- Provided technical leadership across architecture, onboarding, reliability, system troubleshooting, platform integration and developer enablement while continuing to write production code.

Managing Director | TESLA Asia Pacific

Mar 2021 - May 2024 | Auckland, New Zealand

- Led TESLA Forecasting's Asia-Pacific operations across engineering, forecasting analysis, support, sales, business development and customer delivery.
- Built and developed a high-performing 10-person multidisciplinary team supporting APAC energy forecasting customers across multiple functions and countries.
- Combined commercial leadership with hands-on technical contribution, ensuring APAC operations remained closely connected to product capability, customer requirements and operational delivery.
- Managed senior customer and partner relationships across utilities, system operators, retailers, traders and weather/data providers.
- Supported integration into Yes Energy following acquisition while progressively handing over finance, payroll, contracts, HR, banking, invoicing and operational responsibilities to create capacity for Distinguished Engineer work.
- Maintained strong team culture, mentoring, onboarding and operational continuity through organisational change, cross-time-zone collaboration and business integration.

Regional Director | TESLA Asia Pacific

Dec 2010 - Mar 2021 | Auckland, New Zealand

- Established and grew TESLA Forecasting's APAC presence from the ground up, covering operations, customer relationships, business development, product support and regional delivery.
- Worked as the primary senior technical and operational lead in New Zealand, collaborating with global teams across the United States, United Kingdom and Asia-Pacific region.
- Built trusted relationships with energy market customers and partners while supporting forecasting implementations, operational reliability and ongoing product development.
- Served as primary developer of TESLAWeb2 during a substantial period of its evolution, supporting the platform used by customers globally.
- Demonstrated long-term effectiveness as a remote and distributed leader, including extended periods working independently across time zones while delivering critical product and operational outcomes.

Technical Manager | TESLA Europe Limited

Jul 2002 - Nov 2010 | Greater London, United Kingdom

- Led and contributed to development of commercial electricity and gas forecasting software for energy market customers.
- Architected, developed and supported forecasting applications using C++, C#, .NET, SQL Server and web technologies.
- Developed deep domain expertise in energy forecasting, load forecasting, market operations and customer-facing forecasting workflows.
- Worked directly with analysts and customers to translate complex forecasting requirements into software capability and operational processes.
- Supported a forecasting business serving major energy market participants, including clients serving over 85% of the electricity market in England, Wales and Scotland, plus US customers with a combined peak load of over 140GW.

EARLIER CAREER

InstallShield Contractor | Barclays Bank PLC | Jul 2002 - Created installation packages for internal banking applications and Oracle database access components.

InstallShield Contractor | HSBC AssetFinance (UK) Ltd | Dec 2001 - Apr 2002 - Completed approximately 200 software installation packages for remote deployment across standard Windows NT environments.

Software Engineer | FFastFill Plc | Apr 2001 - Nov 2001 - Developed middle-tier trading software and early .NET/C# utilities for electronic trading, futures, options and financial gateway systems.

Software Engineer | Trimble Navigation New Zealand Limited | Jan 1999 - Feb 2001 - Developed C++/COM components, GPS receiver integrations, graphics libraries, technical documentation and public software installers within a shared CORE engineering group.

Assistant System Developer | MAF Quality Management | Dec 1997 - Oct 1998 - Developed Access/SQL database and reporting systems for laboratory proficiency programmes, including SQL queries, interface design, testing and report layouts.

EDUCATION

- Bachelor of Computing and Mathematical Sciences, Computer Software Engineering - University of Waikato, New Zealand, 1995-1999; First Class Honours
- Best Presentation Award in Computing and Mathematical Sciences - University of Waikato, 1998

ADDITIONAL TRAINING & DEVELOPMENT

- Microsoft Foundry / Azure AI Foundry - self-directed learning and proof-of-concept development.
- AI-assisted development, custom agents, RAG systems, vector databases and MCP - practical applied experience.
- Microservices and event-driven architecture - applied through forecasting platform modernisation work.
- Project Planning and Control Techniques.

AVAILABILITY

Remote from New Zealand. 23+ years of experience working across UK, US, NZ and APAC time zones, with a proven ability to collaborate effectively with distributed engineering, product and customer teams. Interested in senior individual contributor roles globally.